

Task 3: Adversity Prediction Model Architecture

Choice of Model and Rationale

To predict the probability of a trade resulting in an adverse outcome (negative PnL) at horizons $\tau \in \{5, 10, 15, 20, 25, 30\}$, the architecture utilizes Scikit-Learn's **HistGradientBoostingClassifier** (an implementation inspired by LightGBM).

Rationale:

- **Suitability for Financial Microstructure Data:** Tree-based ensemble models consistently yield state-of-the-art results on **tabular** financial datasets due to their ability to naturally model highly non-linear relationships and intricate feature interactions (e.g., volume interacting with client directional imbalance).
- **Scale Invariance:** Tree-based models partition features based on threshold rules, making them invariant to monotonic transformations. This eliminates the need for complex feature scaling (e.g., standardizing heavy-tailed variables like trade volume).
- **Evaluation Alignment and Leakage Prevention:** The models were tuned specifically to minimize `neg_log_loss`, directly aligning with probabilistic adversity estimation. To ensure the models generalized safely, hyperparameter search relied strictly on a *TimeSeriesSplit*. This cross-validation technique strictly respects chronological order, guaranteeing no future market data leaked into the training folds. The consistent test/validation log-loss (≈ 0.6844) and stable out-of-sample accuracy affirm the model's robustness against overfitting.

Feature Selection and Justification

A 18-dimensional feature vector was explicitly engineered to capture the underlying drivers of adverse selection without incorporating any look-ahead bias. The features are fundamentally grouped into four categories:

1. Client State & Toxicity Proxies (The "Informed Flow" Detectors)

- *Features:* `client_enc`, `client_avg_volume_20`, `client_volume_ratio`, `client_side_imbalance`, `vol_x_side_imbalance`.
- *Justification:* As observed in Task 1, identifying informed flow is critical to modeling adversity. The `client_side_imbalance` tracks if a client is systematically hitting the bid or lifting the ask over their last 20 trades. The interaction term `vol_x_side_imbalance` acts as a powerful toxicity proxy: if a client executes a trade size far exceeding their moving average (`client_volume_ratio`) in the exact same direction as their recent aggressive flow, it strongly signals informed, directional trading. All client-level rolling metrics strictly use `shift(1)` to prevent data leakage.

2. Market Microstructure & Volatility

- *Features:* `realized_vol`, `ret_5`, `ret_20`.
- *Justification:* Mid-price returns (`ret_5`, `ret_20`) establish local momentum, as trades executed against strong short-term trends are highly likely to become adverse quickly. The `realized_vol` (20-trade RMS mid return) quantifies ambient market instability. High volatility geometrically increases the probability that the mid-price will quickly drift outside the Liquidity Provider's spread boundaries before the position can be safely closed.

3. Order Footprint & Interactions

- *Features:* side, log_volume, spread, relative_spread, side_spread, log_vol_x_rspread.
- *Justification:* log_volume compresses the heavy tails of large block trades. relative_spread normalizes the bid-ask width against the asset’s nominal price. The interaction terms (log_vol_x_rspread) allow the model to specifically isolate scenarios where a massive order consumes liquidity during periods of widened spreads—a classic predecessor to adverse price dislocations.

4. Temporal Constraints

- *Features:* hour, eta.
- *Justification:* eta tracks the elapsed fraction of the trading day. Market microstructure dictates that volatility, spread widths, and informed trading density follow strict intraday U-shaped profiles (high at open and close). The temporal variables anchor the model’s baseline adversity expectation to these macroeconomic daily cycles.

Model Performance Metrics

The table below summarizes the model’s predictive performance averaged across all horizons (τ):

Split	Accuracy	Precision	Recall	Log Loss
Train	0.5605	0.6183	0.2519	0.6832
Validation	0.5581	0.4545	0.2506	0.6844
Test	0.5580	0.4485	0.2484	0.6844

Evaluation Commentary:

- **The Good (Stability & Generalization):** The model exhibits exceptional stability across splits. The training, validation, and test metrics are virtually identical (e.g., test accuracy of 0.5580 vs. validation of 0.5581, and an identical log-loss of 0.6844). This confirms that the model has successfully avoided overfitting and generalizes remarkably well to out-of-sample data.
- **The Bad (Low Recall):** The recall score sits at approximately 25% across the out-of-sample splits. This indicates that the model is highly conservative; it only successfully flags a quarter of the truly adverse trades. While precision is moderate ($\approx 45\%$), the low recall means the Liquidity Provider will still inadvertently internalize a significant amount of toxic flow.